

AI Risk Categorization Decoded (AIR 2024)

This summary covers the **AI Risk Taxonomy (AIR 2024)**, a framework designed to bridge the gap between high-level government regulations and granular corporate usage policies. Grounded in an inductive, bottom-up content analysis of **8 government policies** (from the EU, US, and China) and **16 company policies** across 9 global foundation model developers , the taxonomy establishes a standardized language for generative AI safety and evaluation.

1. Core Structure of the AIR 2024 Taxonomy

The taxonomy structures **314 unique risk categories** into a strict, four-tiered hierarchy designed to capture the full spectrum of AI deployment challenges:

- **Level-1 (4 Categories):** Broad, macro-level domains of risk.
- **Level-2 (16 Categories):** Groupings based on overarching societal or functional impact.
- **Level-3 (45 Categories):** Intermediate conceptual groupings that expand on Level-2 themes.
- **Level-4 (314 Categories):** Granular, highly specific risks explicitly named within analyzed policy texts.

Macro Domains (Level-1) Breakdown

The entire framework is anchored by four key pillars:

Level-1 Domain	Scope & Focus	High-Profile Risk Examples (Level-4)
System & Operational Risks	Cybersecurity, critical infrastructure, and misapplication in regulated fields.	Malware, network intrusion, predictive policing, and automated credit scoring.
Content Safety Risks	Direct harms stemming from generated output; protecting end-users.	CSAM, non-consensual intimate imagery (NCII), cyber-bullying, and bioweapon generation.

Societal Risks	Systemic and macro-structural impacts on politics, psychology, and markets.	Election interference/voter discouragement, deepfake impersonation, and academic plagiarism.
Legal & Rights Risks	Infringements on hard legal protections, civil liberties, and regulatory compliance.	PII theft, algorithmic discrimination (e.g., race, age, gender), and IP/trade secret violations.

2. Private Sector Policy Insights (Corporate Acceptable Use)

The paper isolates corporate policies into two distinct types: **platform-wide acceptable use policies** (e.g., OpenAI, Google, Anthropic web interface terms) and **model-specific licenses** for open-weights models (e.g., Meta's Llama, Google's Gemma).

Key Corporate Trends & Vulnerabilities

- **High Consensus Areas:** Risks like *Unauthorized Privacy Violations*, *Types of Sensitive Data*, *Other Unlawful Activities*, and *Harassment* are universally prohibited across every single reviewed corporate policy.
- **The Content Gap:** Despite high alignment on violent/hateful content, extreme gaps exist for *Non-Consensual Nudity (NCII)* and *Offensive Language*. Stability AI is the sole provider prioritizing text-to-image NCII, while traditional text-based model providers often omit explicit phrasing for it.
- **The Corporate "Liability Shift":** AI developers frequently struggle to specify risks associated with *Autonomous Unsafe Operation of Systems*. Instead of building technical boundaries, companies often use legal disclaimers to push liability entirely onto the end-user.
- **Policy Streamlining:** Over time, pioneers like OpenAI have consolidated highly granular risk lists (like 2023's detailed violence/military breakdown) into simplified, generalized legal categories, shifting focus toward socio-political stabilization and election defense.

3. Public Sector Comparative Analysis (Jurisdictional Regimes)

A core contribution of the study is mapping public-sector frameworks across the **European Union, United States, and China**, revealing stark ideological and structural divergences.

A. European Union: The Rigorous, Human-Centric Tier

- **Primary Instruments:** General Data Protection Regulation (GDPR) and the EU AI Act.
- **Focus:** Strong emphasis on the protection of fundamental human rights and vulnerable groups.
- **Binding vs. Non-binding Gap:** While the EU discusses **23 Level-3 risks** across its legal recitals (preambles), only **7 Level-3 categories** carry mandatory, hard compliance obligations for high-risk systems (e.g., biometric profiling, automated employment screening).
- **EU Unique Feature:** The EU AI Act uniquely prohibits systems that intentionally exploit human vulnerabilities related to *age, economic standing, or social conditions*.

B. United States: The Labor and Market Sentinel

- **Primary Instrument:** White House Executive Order (EO) on Safe, Secure, and Trustworthy AI (drawing heavily from the voluntary NIST Risk Management Framework).
- **Focus:** National security, algorithmic bias in civic spaces, and market competitiveness.
- **US Unique Feature:** The US document is the only regulatory text that details the **Disempowering/Displacing Workers** category. It explicitly targets risks regarding *undue algorithmic worker surveillance, worsening job quality, and economic labor-force disruptions*.

C. China: The Strict Social Stability and Regime Security Paradigm

- **Primary Instruments:** A matrix of targeted rules including the *Generative AI Services Measures* and *Basic Security Requirements*.
- **Focus:** Extreme focus on content filtering, algorithmic accountability, and ideological continuity.
- **China Unique Features:** Features multiple "China-Unique" risk classifications, including *Disrupting Social Order, Subverting State Power, Undermining National Unity*, and specific protections for *Image Rights/Dignity/Honor*.
- **Compliance Reality:** Domestically headquartered firms (e.g., DeepSeek and Baidu) achieve a staggering **90–95% direct text alignment** with government mandates, reflecting the strict licensing and red-teaming enforcement of Chinese authorities.

4. Key Takeaways for Legal and Policy Analysis

For a department focused on the legal dimensions of AI, the paper delivers three critical conclusions:

1. **Regulatory Under-Specification:** Despite political rhetoric claiming government frameworks are over-regulated, a hard textual audit reveals that **corporate acceptable use policies are vastly more granular and restrictive** than public laws. Governments have massive room to codify existing corporate safety baselines into law without increasing compliance strain on major companies.
2. **Superficial Alignment Flaws:** While international bodies frequently agree on broad Level-2 risk labels (e.g., "Deception" or "Operational Misuse"), an audit of their respective Level-4 entries reveals **zero** practical convergence. True global harmony or interoperability in AI law requires aligning definitions at the lower-tier, operational level.
3. **The Labor Governance Vacuum:** The private sector completely ignores labor harms. No reviewed corporate policy provides guardrails against the automated exploitation, deskilling, or hyper-surveillance of workforce environments, leaving a massive regulatory gap that currently rests solely on US Executive Order directives.